Since radius does not change with time

$$\dot{r}_T = 0 \quad \& \quad \ddot{r}_T = 0$$

From Keplerian motion identity,

$$\ddot{r}_T = \dot{\theta}^2 r_T - \frac{\mu}{r_T^2}$$

Substituting,

$$0 = \dot{\theta}^2 a - \frac{\mu}{a^2} \Rightarrow \dot{\theta}^2 = \frac{\mu}{a^3}$$

$$\Rightarrow \dot{\theta} = n = \sqrt{\frac{\mu}{a^3}} \quad (n = \text{mean motion})$$

Then $\ddot{\theta}$:

From Keplerian identity

$$\ddot{\theta} = -2\frac{\dot{r}_T}{r_T}\dot{\theta}$$

$$\Rightarrow \ddot{\theta} = 0 \quad (\because \dot{r}_T = 0)$$

∴ For mean motion,

$$\frac{\mu}{r_T^3} = \frac{\mu}{a^3} = n^2 = \dot{\theta}^2$$

⇒ Radial (x):

$$\ddot{x} - x(n^2 + 2n^2) - 2n\dot{y} = 0$$

$$\Rightarrow \boxed{\ddot{x} - 3n^2 x - 2n\dot{y} = 0}$$

Along track (y):

$$\ddot{y} + 2n\dot{x} - y(n^2 - n^2) = 0$$

$$\Rightarrow \boxed{\ddot{y} + 2n\dot{x} = 0}$$

And cross track (z):

$$\boxed{\ddot{z} + n^2 z = 0}$$

These are Hill-Clohessy-Wiltshire (HCW) equations.

(5) Integrate the along track equation & solve for $x(t), y(t)$

i) along track eq.

$$\ddot{y} + 2n\dot{x} = 0$$

Integrating once

$$\dot{y} = -2nx + d$$

$$\Rightarrow d = \dot{y}_0 + 2nx_0$$

d ← integration constant

↳ substitute into the radial equation

$$\ddot{x} - 3n^2x - 2n(-2nx + d) = 0$$

Simplifying,

$$\ddot{x} + n^2x = 2nd$$ → forced harmonic oscillator

∃ Homogeneous equation

$$\ddot{x} + n^2x = 0$$

$$x_h(t) = C_1 \cos(nt) + C_2 \sin(nt)$$

Particular solution

$$x_p = C_3 \Rightarrow \ddot{x}_p = 0$$

$$n^2 x_p = C_1 \cos(nt) + C_2 \sin(nt) + \frac{2d}{h}$$

applying initial conditions

at $t = 0$, $x(0) = x_0$, $\dot{x}(0) = \dot{x}_0$

$$x_0 = C_1 + \frac{2d}{h} \Rightarrow C_1 = x_0 - \frac{2d}{h}$$

Differentiating

$$\dot{x}(t) = -C_1 n \sin(nt) + C_2 n \cos(nt)$$

at $t = 0$,

$$\dot{x}_0 = C_2 n \Rightarrow C_2 = \frac{\dot{x}_0}{n}$$

Subbing in C_1 & C_2

$$x(t) = \left(x_0 - \frac{2d}{n}\right)\cos(nt) + \frac{\dot{x}_0}{n}\sin(nt) + \frac{2d}{n}$$

Using $d = \dot{y}_0 + 2nx_0$

$$\Rightarrow \boxed{x(t) = \frac{\dot{x}_0}{n}\sin(nt) - \left(3x_0 + \frac{2\dot{y}_0}{n}\right)\cos(nt) + \left(4x_0 + \frac{2\dot{y}_0}{n}\right)}$$

Now solving for $y(t)$

From along track equation

$$\dot{y} = -2nx + d$$

substituting expression for $x(t)$:

$$\dot{y} = -2nC_1\cos(nt) - 2nC_2\sin(nt) - 3d$$

Integrating

$$y(t) = -2C_1\sin(nt) + 2C_2\cos(nt) - 3dt + C_4$$

At $t=0$, $y(0)=y_0$

$$y_0 = 2C_2 + C_4$$

$$C_4 = y_0 - \frac{2\dot{x}_0}{n}$$

Substituting C_1, C_2, d, C_4

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{n}\right)\sin(nt) + \frac{2\dot{x}_0}{n}\cos(nt) - (6nx_0 + 3\dot{y}_0)\,t + \left(y_0 - \frac{2\dot{x}_0}{n}\right)$$

© Secular term and bounded motion condition

From the along-track solution above we have

$$y(t) = \text{.----.}$$

The sinusoidal terms are periodic & bounded

The term that grows with time is the secular (drifting) term:

$$y_{secular}(t) = -(6nx_0 + 3\dot{y}_0)t$$

For bounded relative motion, the secular drift must be zero:

$$6nx_0 + 3\dot{y}_0 = 0$$

$$3\dot{y}_0 = -6nx_0$$

$\therefore$ Bounded motion condition is

$$\boxed{\dot{y}_0 = -2nx_0}$$

(d) Show equivalence to $\Delta a = 0$

From circular target orbit element mapping

$$x_0 = \Delta a - a\Delta e \cos\theta_0$$

$$\dot{y}_0 = -\frac{3n}{2}\Delta a + 2nc\Delta e \cos\theta_0$$

From initial radial position;

$$a\Delta e \cos\theta_0 = \Delta a - x_0$$

→ Put into $\dot{y}_0$

$$\dot{y}_0 = -\frac{3n}{2}\Delta a + 2n(\Delta a - x_0)$$

$$\Rightarrow \dot{y}_0 = -\frac{3n}{2}\Delta a + 2n\Delta a - 2n x_0$$

$$\Rightarrow \dot{y}_0 = \frac{n}{2}\Delta a - 2n x_0$$

$$\Rightarrow \boxed{\dot{y}_0 + 2n x_0 = \frac{n}{2}\Delta a}$$

From part (c) bounded motion condition

$$\dot{y}_0 + 2nx_0 = 0$$

$$\Rightarrow \frac{n}{2}\Delta a = 0$$

$$\because n \neq 0 \Rightarrow \boxed{\Delta a = 0}$$

∴ The two conditions are equivalent

Why is orbit-element condition more general?

The Cartesian condition $\dot{y}_0 = -2nx_0$ is obtained from HCW equations which assume a circular target orbit whereas the orbit-element condition $\Delta a = 0$ means the s/c have equal semi-major axes.

$\because n = \sqrt{\frac{\mu}{a^3}}$ they have equal mean motions and hence orbital periods $\left(\text{as } n = \frac{2\pi}{T}\right)$ under unperturbed 2-body motion, preventing secular along-track drift

This also applies to eccentric Keplerian target orbits, whereas the stated HCW condition is specific to circular target approximation.